



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Evaluate limits by direct substitution
- Use factoring to resolve 0/0 indeterminate forms
- Apply one-sided and two-sided limit definitions
- Determine where a function is continuous

Simplify each expression completely. Show all steps and circle your final answer.

Limits by factoring

1. Evaluate: $\lim$ as x approaches 1 of $[1(x^2 - 1^2) / (x - 1)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 1} \frac{1(x^2 - 1^2)}{x - 1}$$

Answer: _____

2. Evaluate: $\lim$ as x approaches 4 of $[1(x^2 - 4^2) / (x - 4)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 4} \frac{1(x^2 - 4^2)}{x - 4}$$

Answer: _____

3. Evaluate: $\lim$ as x approaches 3 of $[4(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{4(x^2 - 3^2)}{x - 3}$$

Answer: _____

4. Evaluate: $\lim$ as x approaches 4 of $[2(x^2 - 4^2) / (x - 4)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 4} \frac{2(x^2 - 4^2)}{x - 4}$$

Answer: _____

5. Evaluate: $\lim$ as x approaches 2 of $[2(x^2 - 2^2) / (x - 2)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 2} \frac{2(x^2 - 2^2)}{x - 2}$$

Answer: _____



6. Evaluate: $\lim$ as x approaches 1 of $[2(x^2 - 1^2) / (x - 1)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 1} \frac{2(x^2 - 1^2)}{x - 1}$$

Answer: _____

7. Evaluate: $\lim$ as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

Answer: _____

8. Evaluate: $\lim$ as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

Answer: _____

Limits by direct substitution

9. Evaluate the limit by direct substitution: $\lim$ (x to 1) of $(1x^2 + 0x + -3)$.

$$\lim_{x \rightarrow 1} (1x^2 + 0x - 3)$$

Answer: _____

10. A price-demand function is $p(x) = -1x^2 + 13x + 119$. Find the limit as x approaches 2 and interpret it as the price when demand approaches 2 units.

$$\lim_{x \rightarrow 2} (-1x^2 + 13x + 119)$$

Answer: _____

11. Evaluate: limit as x approaches 1 of $(2x^2 + 0x + 7)$.

$$\lim_{x \rightarrow 1} (2x^2 + 0x + 7)$$

Answer: _____

12. Evaluate the limit by direct substitution: $\lim$ (x to 2) of $(3x^2 + -3x + 1)$.

$$\lim_{x \rightarrow 2} (3x^2 - 3x + 1)$$

Answer: _____



13. A price-demand function is $p(x) = -1x^2 + 7x + 121$. Find the limit as x approaches 4 and interpret it as the price when demand approaches 4 units.

$$\lim_{x \rightarrow 4} (-1x^2 + 7x + 121)$$

Answer: _____

14. Evaluate: limit as x approaches -2 of $(1x^2 + -1x + 5)$.

$$\lim_{x \rightarrow -2} (1x^2 - 1x + 5)$$

Answer: _____

15. Evaluate the limit by direct substitution: $\lim (x \text{ to } -1)$ of $(2x^2 + 4x + 3)$.

$$\lim_{x \rightarrow -1} (2x^2 + 4x + 3)$$

Answer: _____

16. A price-demand function is $p(x) = -1x^2 + 5x + 142$. Find the limit as x approaches 5 and interpret it as the price when demand approaches 5 units.

$$\lim_{x \rightarrow 5} (-1x^2 + 5x + 142)$$

Answer: _____

17. Evaluate: limit as x approaches 4 of $(3x^2 + -3x + 6)$.

$$\lim_{x \rightarrow 4} (3x^2 - 3x + 6)$$

Answer: _____

18. Evaluate the limit by direct substitution: $\lim (x \text{ to } 3)$ of $(1x^2 + -1x + -3)$.

$$\lim_{x \rightarrow 3} (1x^2 - 1x - 3)$$

Answer: _____

19. A price-demand function is $p(x) = -2x^2 + 12x + 133$. Find the limit as x approaches 4 and interpret it as the price when demand approaches 4 units.

$$\lim_{x \rightarrow 4} (-2x^2 + 12x + 133)$$

Answer: _____



20. Evaluate: limit as x approaches 0 of $(1x^2 + -1x + -8)$.

$$\lim_{x \rightarrow 0} (1x^2 - 1x - 8)$$

Answer: _____

21. Evaluate the limit by direct substitution: $\lim (x \text{ to } 1)$ of $(2x^2 + -3x + 5)$.

$$\lim_{x \rightarrow 1} (2x^2 - 3x + 5)$$

Answer: _____

22. A price-demand function is $p(x) = -1x^2 + 9x + 123$. Find the limit as x approaches 3 and interpret it as the price when demand approaches 3 units.

$$\lim_{x \rightarrow 3} (-1x^2 + 9x + 123)$$

Answer: _____

23. Evaluate: limit as x approaches 2 of $(2x^2 + -4x + -2)$.

$$\lim_{x \rightarrow 2} (2x^2 - 4x - 2)$$

Answer: _____

24. Evaluate the limit by direct substitution: $\lim (x \text{ to } 0)$ of $(4x^2 + 3x + -1)$.

$$\lim_{x \rightarrow 0} (4x^2 + 3x - 1)$$

Answer: _____

25. A price-demand function is $p(x) = -1x^2 + 6x + 130$. Find the limit as x approaches 5 and interpret it as the price when demand approaches 5 units.

$$\lim_{x \rightarrow 5} (-1x^2 + 6x + 130)$$

Answer: _____

26. Evaluate: limit as x approaches 0 of $(2x^2 + -2x + 1)$.

$$\lim_{x \rightarrow 0} (2x^2 - 2x + 1)$$

Answer: _____

27. Evaluate the limit by direct substitution: $\lim (x \text{ to } 1)$ of $(1x^2 + -3x + 1)$.

$$\lim_{x \rightarrow 1} (1x^2 - 3x + 1)$$

Answer: _____



28. A price-demand function is $p(x) = -1x^2 + 13x + 64$. Find the limit as x approaches 3 and interpret it as the price when demand approaches 3 units.

$$\lim_{x \rightarrow 3} (-1x^2 + 13x + 64)$$

Answer: _____

29. Evaluate: limit as x approaches -1 of $(3x^2 + -6x + -5)$.

$$\lim_{x \rightarrow -1} (3x^2 - 6x - 5)$$

Answer: _____

30. Evaluate the limit by direct substitution: $\lim_{x \rightarrow -2} (3x^2 + 4x + 4)$.

$$\lim_{x \rightarrow -2} (3x^2 + 4x + 4)$$

Answer: _____





MATH120: Limits

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ANSWER KEY & SOLUTIONS

Topics: Limits by direct substitution, Limits by factoring. All answers verified by independent computation.

Solutions



Limits by factoring

1. Evaluate: $\lim$ as x approaches 1 of $[1(x^2 - 1^2) / (x - 1)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 1} \frac{1(x^2 - 1^2)}{x - 1}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 1$: $1^*(1+1) = 1*2 = 2$.

Answer: $= (1)(x + 1)|_{x=1} = 2$

2. Evaluate: $\lim$ as x approaches 4 of $[1(x^2 - 4^2) / (x - 4)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 4} \frac{1(x^2 - 4^2)}{x - 4}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 4$: $1^*(4+4) = 1*2*4 = 8$.

Answer: $= (1)(x + 4)|_{x=4} = 8$

3. Evaluate: $\lim$ as x approaches 3 of $[4(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{4(x^2 - 3^2)}{x - 3}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 3$: $4^*(3+3) = 4*2*3 = 24$.

Answer: $= (4)(x + 3)|_{x=3} = 24$

4. Evaluate: $\lim$ as x approaches 4 of $[2(x^2 - 4^2) / (x - 4)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 4} \frac{2(x^2 - 4^2)}{x - 4}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 4$: $2^*(4+4) = 2*2*4 = 16$.

Answer: $= (2)(x + 4)|_{x=4} = 16$

5. Evaluate: $\lim$ as x approaches 2 of $[2(x^2 - 2^2) / (x - 2)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 2} \frac{2(x^2 - 2^2)}{x - 2}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 2$: $2^*(2+2) = 2*2*2 = 8$.

Answer: $= (2)(x + 2)|_{x=2} = 8$



6. Evaluate: $\lim$ as x approaches 1 of $[2(x^2 - 1^2) / (x - 1)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 1} \frac{2(x^2 - 1^2)}{x - 1}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 1$: $2^*(1+1) = 2*2*1 = 4$.

Answer: $= (2)(x + 1)|_{x=1} = 4$

7. Evaluate: $\lim$ as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 3$: $1^*(3+3) = 1*2*3 = 6$.

Answer: $= (1)(x + 3)|_{x=3} = 6$

8. Evaluate: $\lim$ as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 form — factor first.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

→ Factor the numerator: $a^*(x^2 - r^2) = a^*(x-r)^*(x+r)$.

→ Cancel $(x-r)$ top and bottom. The limit simplifies to $a^*(x+r)$.

→ Substitute $x = 3$: $1^*(3+3) = 1*2*3 = 6$.

Answer: $= (1)(x + 3)|_{x=3} = 6$



Limits by direct substitution

9. Evaluate the limit by direct substitution: $\lim_{x \rightarrow 1} (1x^2 + 0x + -3)$.

$$\lim_{x \rightarrow 1} (1x^2 + 0x - 3)$$

→ Since the function is a polynomial, substitute $x = 1$ directly.

$$\rightarrow = 1(1)^2 + 0(1) + -3 = 1 + 0 + -3 = -2.$$

Answer: $\lim_{x \rightarrow 1} = 1 + 0 + -3 = -2$

10. A price-demand function is $p(x) = -1x^2 + 13x + 119$. Find the limit as x approaches 2 and interpret it as the price when demand approaches 2 units.

$$\lim_{x \rightarrow 2} (-1x^2 + 13x + 119)$$

→ Substitute $x = 2$: $\lim = -1(2)^2 + 13(2) + 119$.

$$\rightarrow = -4 + 26 + 119 = 141.$$

→ The price approaches \$141 as demand approaches 2 units.

Answer: $\lim_{x \rightarrow 2} = -4 + 26 + 119 = 141$

11. Evaluate: limit as x approaches 1 of $(2x^2 + 0x + 7)$.

$$\lim_{x \rightarrow 1} (2x^2 + 0x + 7)$$

→ Polynomial limit — plug in $x = 1$ directly.

$$\rightarrow = 2(1)^2 + 0(1) + 7 = 9.$$

Answer: $\lim_{x \rightarrow 1} = 2 + 0 + 7 = 9$

12. Evaluate the limit by direct substitution: $\lim_{x \rightarrow 2} (3x^2 + -3x + 1)$.

$$\lim_{x \rightarrow 2} (3x^2 - 3x + 1)$$

→ Since the function is a polynomial, substitute $x = 2$ directly.

$$\rightarrow = 3(2)^2 + -3(2) + 1 = 12 + -6 + 1 = 7.$$

Answer: $\lim_{x \rightarrow 2} = 12 + -6 + 1 = 7$

13. A price-demand function is $p(x) = -1x^2 + 7x + 121$. Find the limit as x approaches 4 and interpret it as the price when demand approaches 4 units.

$$\lim_{x \rightarrow 4} (-1x^2 + 7x + 121)$$

→ Substitute $x = 4$: $\lim = -1(4)^2 + 7(4) + 121$.

$$\rightarrow = -16 + 28 + 121 = 133.$$

→ The price approaches \$133 as demand approaches 4 units.

Answer: $\lim_{x \rightarrow 4} = -16 + 28 + 121 = 133$



14. Evaluate: limit as x approaches -2 of $(1x^2 + -1x + 5)$.

$$\lim_{x \rightarrow -2} (1x^2 - 1x + 5)$$

→ Polynomial limit — plug in $x = -2$ directly.

$$\rightarrow = 1(-2)^2 + -1(-2) + 5 = 11.$$

Answer: $\lim_{x \rightarrow -2} = 4 + 2 + 5 = 11$

15. Evaluate the limit by direct substitution: $\lim (x \text{ to } -1) \text{ of } (2x^2 + 4x + 3)$.

$$\lim_{x \rightarrow -1} (2x^2 + 4x + 3)$$

→ Since the function is a polynomial, substitute $x = -1$ directly.

$$\rightarrow = 2(-1)^2 + 4(-1) + 3 = 2 + -4 + 3 = 1.$$

Answer: $\lim_{x \rightarrow -1} = 2 + -4 + 3 = 1$

16. A price-demand function is $p(x) = -1x^2 + 5x + 142$. Find the limit as x approaches 5 and interpret it as the price when demand approaches 5 units.

$$\lim_{x \rightarrow 5} (-1x^2 + 5x + 142)$$

→ Substitute $x = 5$: $\lim = -1(5)^2 + 5(5) + 142$.

$$\rightarrow = -25 + 25 + 142 = 142.$$

→ The price approaches \$142 as demand approaches 5 units.

Answer: $\lim_{x \rightarrow 5} = -25 + 25 + 142 = 142$

17. Evaluate: limit as x approaches 4 of $(3x^2 + -3x + 6)$.

$$\lim_{x \rightarrow 4} (3x^2 - 3x + 6)$$

→ Polynomial limit — plug in $x = 4$ directly.

$$\rightarrow = 3(4)^2 + -3(4) + 6 = 42.$$

Answer: $\lim_{x \rightarrow 4} = 48 + -12 + 6 = 42$

18. Evaluate the limit by direct substitution: $\lim (x \text{ to } 3) \text{ of } (1x^2 + -1x + -3)$.

$$\lim_{x \rightarrow 3} (1x^2 - 1x - 3)$$

→ Since the function is a polynomial, substitute $x = 3$ directly.

$$\rightarrow = 1(3)^2 + -1(3) + -3 = 9 + -3 + -3 = 3.$$

Answer: $\lim_{x \rightarrow 3} = 9 + -3 + -3 = 3$



19. A price-demand function is $p(x) = -2x^2 + 12x + 133$. Find the limit as x approaches 4 and interpret it as the price when demand approaches 4 units.

$$\lim_{x \rightarrow 4} (-2x^2 + 12x + 133)$$

→ Substitute $x = 4$: $\lim = -2(4)^2 + 12(4) + 133$.

→ $= -32 + 48 + 133 = 149$.

→ The price approaches \$149 as demand approaches 4 units.

Answer: $\lim_{x \rightarrow 4} = -32 + 48 + 133 = 149$

20. Evaluate: limit as x approaches 0 of $(1x^2 + -1x + -8)$.

$$\lim_{x \rightarrow 0} (1x^2 - 1x - 8)$$

→ Polynomial limit — plug in $x = 0$ directly.

→ $= 1(0)^2 + -1(0) + -8 = -8$.

Answer: $\lim_{x \rightarrow 0} = 0 + 0 + -8 = -8$

21. Evaluate the limit by direct substitution: $\lim_{x \rightarrow 1}$ of $(2x^2 + -3x + 5)$.

$$\lim_{x \rightarrow 1} (2x^2 - 3x + 5)$$

→ Since the function is a polynomial, substitute $x = 1$ directly.

→ $= 2(1)^2 + -3(1) + 5 = 2 + -3 + 5 = 4$.

Answer: $\lim_{x \rightarrow 1} = 2 + -3 + 5 = 4$

22. A price-demand function is $p(x) = -1x^2 + 9x + 123$. Find the limit as x approaches 3 and interpret it as the price when demand approaches 3 units.

$$\lim_{x \rightarrow 3} (-1x^2 + 9x + 123)$$

→ Substitute $x = 3$: $\lim = -1(3)^2 + 9(3) + 123$.

→ $= -9 + 27 + 123 = 141$.

→ The price approaches \$141 as demand approaches 3 units.

Answer: $\lim_{x \rightarrow 3} = -9 + 27 + 123 = 141$

23. Evaluate: limit as x approaches 2 of $(2x^2 + -4x + -2)$.

$$\lim_{x \rightarrow 2} (2x^2 - 4x - 2)$$

→ Polynomial limit — plug in $x = 2$ directly.

→ $= 2(2)^2 + -4(2) + -2 = -2$.

Answer: $\lim_{x \rightarrow 2} = 8 + -8 + -2 = -2$



24. Evaluate the limit by direct substitution: $\lim_{x \rightarrow 0}$ of $(4x^2 + 3x + -1)$.

$$\lim_{x \rightarrow 0} (4x^2 + 3x - 1)$$

→ Since the function is a polynomial, substitute $x = 0$ directly.

$$\rightarrow = 4(0)^2 + 3(0) + -1 = 0 + 0 + -1 = -1.$$

Answer: $\lim_{x \rightarrow 0} = 0 + 0 + -1 = -1$

25. A price-demand function is $p(x) = -1x^2 + 6x + 130$. Find the limit as x approaches 5 and interpret it as the price when demand approaches 5 units.

$$\lim_{x \rightarrow 5} (-1x^2 + 6x + 130)$$

→ Substitute $x = 5$: $\lim = -1(5)^2 + 6(5) + 130$.

$$\rightarrow = -25 + 30 + 130 = 135.$$

→ The price approaches \$135 as demand approaches 5 units.

Answer: $\lim_{x \rightarrow 5} = -25 + 30 + 130 = 135$

26. Evaluate: limit as x approaches 0 of $(2x^2 + -2x + 1)$.

$$\lim_{x \rightarrow 0} (2x^2 - 2x + 1)$$

→ Polynomial limit — plug in $x = 0$ directly.

$$\rightarrow = 2(0)^2 + -2(0) + 1 = 1.$$

Answer: $\lim_{x \rightarrow 0} = 0 + 0 + 1 = 1$

27. Evaluate the limit by direct substitution: $\lim_{x \rightarrow 1}$ of $(1x^2 + -3x + 1)$.

$$\lim_{x \rightarrow 1} (1x^2 - 3x + 1)$$

→ Since the function is a polynomial, substitute $x = 1$ directly.

$$\rightarrow = 1(1)^2 + -3(1) + 1 = 1 + -3 + 1 = -1.$$

Answer: $\lim_{x \rightarrow 1} = 1 + -3 + 1 = -1$

28. A price-demand function is $p(x) = -1x^2 + 13x + 64$. Find the limit as x approaches 3 and interpret it as the price when demand approaches 3 units.

$$\lim_{x \rightarrow 3} (-1x^2 + 13x + 64)$$

→ Substitute $x = 3$: $\lim = -1(3)^2 + 13(3) + 64$.

$$\rightarrow = -9 + 39 + 64 = 94.$$

→ The price approaches \$94 as demand approaches 3 units.

Answer: $\lim_{x \rightarrow 3} = -9 + 39 + 64 = 94$



29. Evaluate: limit as x approaches -1 of $(3x^2 + -6x + -5)$.

$$\lim_{x \rightarrow -1} (3x^2 - 6x - 5)$$

→ Polynomial limit — plug in $x = -1$ directly.

$$\rightarrow = 3(-1)^2 + -6(-1) + -5 = 4.$$

Answer: $\lim_{x \rightarrow -1} = 3 + 6 + -5 = 4$

30. Evaluate the limit by direct substitution: $\lim_{x \rightarrow -2} (3x^2 + 4x + 4)$.

$$\lim_{x \rightarrow -2} (3x^2 + 4x + 4)$$

→ Since the function is a polynomial, substitute $x = -2$ directly.

$$\rightarrow = 3(-2)^2 + 4(-2) + 4 = 12 + -8 + 4 = 8.$$

Answer: $\lim_{x \rightarrow -2} = 12 + -8 + 4 = 8$

